

Chemistry

Unit 4

Practice Exam 1

This sample test has been prepared as part of the Pearson suite of resources for the Year 12, Unit 4, ATAR Chemistry Course prescribed by the Western Australian School Curriculum and Standard Authority.

Time allowed

Reading time: 10 minutes Working time: 90 minutes

Materials required

An approved non-programmable calculator.

Chemistry Data Sheet. This may be downloaded from the WACE website.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of total exam
Section one: Multiple-choice	16	16	12	32	26
Section two: Short answer	7	7	16	50	42
Section three: Extended answer	2	2	17	38	32
				Total	100

Section one: Multiple-choice

26% (32 marks)

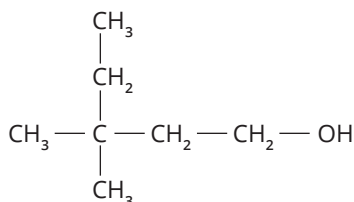
This section has 16 questions. Answer **all** questions by circling the correct option. Do not erase or use correct fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 12 minutes

1 Which one of the following is a correct IUPAC name of an organic compound?

- a cis-1,2-dichloroethane
- b 3-ethylbutan-2-ol
- c methylpropene
- d butan-1-one

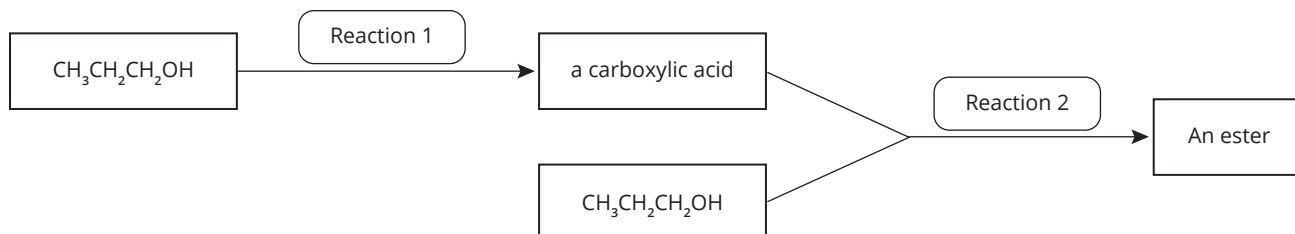
2 What is the systematic name for the following compound?



- a 3-ethyl-3,3-dimethylpropan-1-ol
 - b 3-ethyl-3-methylbutan-1-ol
 - c 2-ethyl-2-methylbutan-4-ol
 - d 3,3-dimethylpentan-1-ol
- 3 Several drops of acidified potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) are added to 5 mL of propanone (CH_3COCH_3). What is the colour of the resultant solution?
- a orange
 - b green
 - c yellow
 - d colourless
- 4 Which one of the following alternatives lists compounds in order of increasing boiling points?
- a propane, butane, butanone, butan-1-ol
 - b butane, butanone, butan-1-ol, propane
 - c butanone, propane, butane, butan-1-ol
 - d propane, butane, butan-1-ol, butanone
- 5 The compounds ethanol ($\text{C}_2\text{H}_5\text{OH}$), ethanoic acid (CH_3COOH) and ethanamine ($\text{C}_2\text{H}_5\text{NH}_2$) are soluble in water. Which one of the following alternatives correctly compares the pH of solutions of these compounds?
- a The pH of all three solutions is the same.
 - b Ethanoic acid and ethanamine have a lower pH than ethanol.
 - c The pH of ethanamine and ethanol are the same as and higher than that of ethanoic acid.
 - d The pH of ethanol is greater than that of ethanoic acid but lower than that of ethanamine.

- 6 Which one of the following compounds is least likely to be one of the amino acids that undergo a condensation reaction to form a protein?
- $\text{H}_2\text{NCH}(\text{CH}_2\text{OH})\text{COOH}$
 - $\text{H}_2\text{NCH}(\text{CH}_2\text{COOH})\text{COOH}$
 - $\text{H}_2\text{NCH}_2\text{CH}(\text{CH}_2\text{SH})\text{COOH}$
 - $\text{H}_2\text{NCH}(\text{CH}_2\text{CH}_2\text{CONH}_2)\text{COOH}$
- 7 Which of the following is the most likely form of the amino acid, glycine, in solution at pH 9?
- $\text{H}_2\text{NCH}_2\text{COOH}$
 - $\text{H}_2\text{NCH}_2\text{COO}^-$
 - $^+\text{H}_3\text{NCH}_2\text{COOH}$
 - $^+\text{H}_3\text{NCH}_2\text{COO}^-$
- 8 What maintains the secondary, β -pleated sheet structure of a protein?
- peptide links between amino acid residues within a polypeptide chain
 - hydrogen bonds between amide and carbonyl functional groups within a polypeptide chain
 - hydrogen bonds between amide and carbonyl functional groups in adjacent polypeptide chains
 - hydrogen bonds between highly polar groups on the side chains of amino acid residues in adjacent polypeptide chains
- 9 When ethene reacts with steam in the presence of a catalyst, what is the molecular formula of the product formed?
- C_2H_6
 - $\text{C}_2\text{H}_5\text{O}$
 - $\text{C}_2\text{H}_6\text{O}$
 - $\text{C}_2\text{H}_4\text{O}_2$
- 10 Large quantities of hydrogen gas are produced industrially by the steam reforming of natural gas. The equation for this reaction is as follows.
- $$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g}); \quad \Delta H = +206 \text{ kJ mol}^{-1}$$
- Which of the following changes will increase the amount of $\text{H}_2(\text{g})$ at equilibrium?
- increasing temperature
 - increasing the pressure
 - adding a catalyst
- I only
 - II only
 - I and II only
 - I, II and III

- 11** How many different tripeptides can be formed using only the amino acids glycine (Gly) and alanine (Ala)?
- 2
 - 3
 - 6
 - 8
- 12** The following reaction sequence shows the formation of an ester, starting with a primary alcohol.



Which one of the following alternatives correctly identifies the type of reactions and the name of the ester formed?

	Reaction 1	Reaction 2	Name of ester
a	addition	addition	propyl ethanoate
b	oxidation	condensation	propyl propanoate
c	oxidation	esterification	ethyl ethanoate
d	addition	condensation	propyl propanoate

- 13** A newly developed drug, X, is prepared in a difficult three-step reaction sequence. The percentage yield for each step of the sequence is shown below.

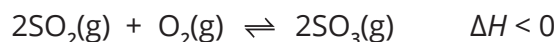
	A	→	B	→	C	→	drug X
% yield		18%		70%		10%	

Which of the following is closest to the overall percentage yield for the conversion of A to drug X?

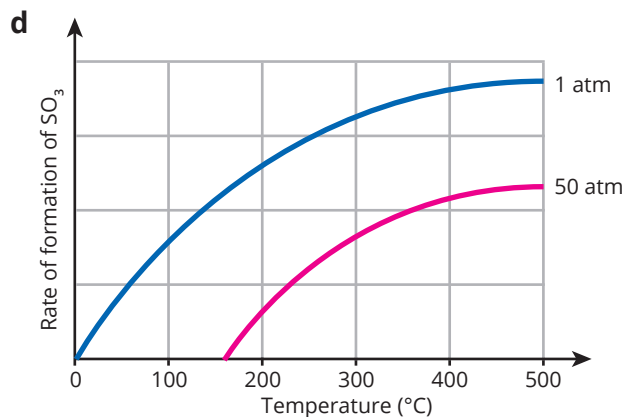
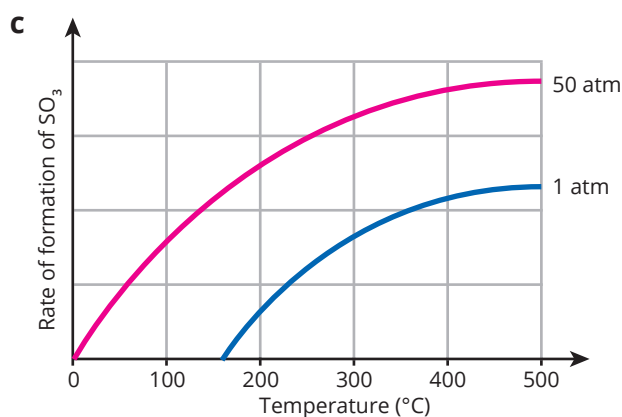
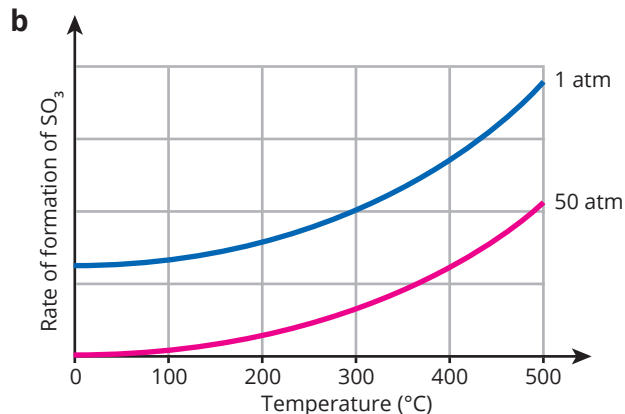
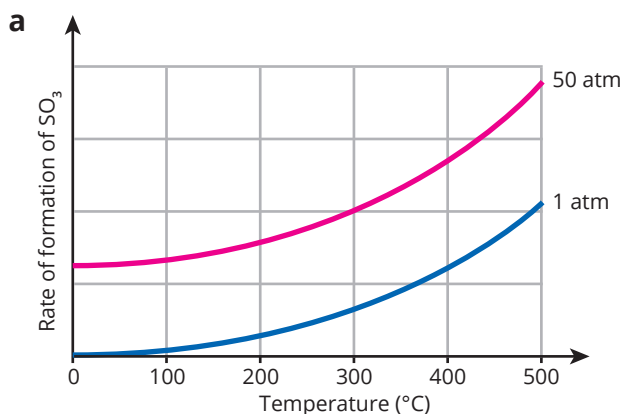
- 1
- 10
- 33
- 98

Questions 14 and 15 refer to the following information.

In the manufacture of sulphuric acid by the Contact Process, sulfur dioxide is catalytically oxidised to sulfur trioxide according to the following equation.



- 14** Which one of the following graphs best shows the effect of increasing the temperature and pressure on the rate of $\text{SO}_3(\text{g})$ formation?



- 15** The actual pressure used in industry for this oxidative step is about 1 atm. What is the reason for this?
- A higher pressure affects the effectiveness of the catalyst.
 - At pressures higher than 1 atm, the equilibrium yield of SO_3 is reduced.
 - At pressures higher than 1 atm, the SO_3 produced would start to decompose.
 - Nearly complete conversion to SO_3 is achieved without incurring the added costs of increasing pressure.
- 16** What group of organic compounds must contain at least three carbon atoms?
- alkenes
 - ketones
 - esters
 - aldehydes
- I only
 - II only
 - II and III only
 - II, III and IV only

End of section one

Section two: Short answer

40% (50 marks)

This section has 7 questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

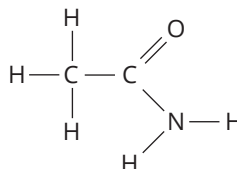
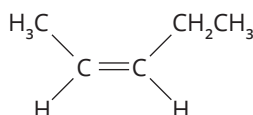
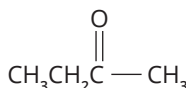
Suggested working time: 16 minutes

Question 17

(9 marks)

- a** Give the IUPAC name for each of the following compounds

(3 marks)



- b** Draw the structural formulas for the following compounds.

(2 marks)

A secondary alcohol with molecular formula $\text{C}_4\text{H}_{10}\text{O}$

An aldehyde with a total of three carbon atoms

- c** Write an equation for the reaction between propan-1-ol and ethanoic acid in the presence of an acid catalyst. Clearly show the structure of the functional group formed in the organic product and give the name of the organic product. (4 marks)

Name of the organic product _____

Question 18**(12 marks)**

Describe two simple chemical tests that could be performed in order to differentiate between members of the following pairs. In each case, include what you would expect to observe and a relevant chemical equation.

a**(6 marks)**

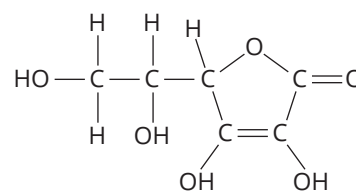
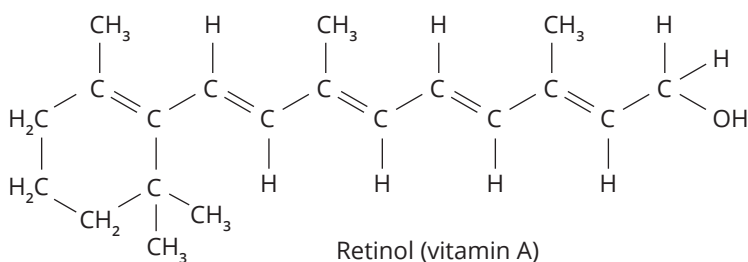
Propan-1-ol and propanoic acid	Test 1
	Observations
	Equation
	Test 2
	Observations
	Equation

b**(6 marks)**

Propan-1-amine and hex-1-ene	Test 1
	Observations
	Equation
	Test 2
	Observations
	Equation

Question 19**(4 marks)**

The following formulas represent two vitamins essential in human nutrition.



a Name two functional groups common to these two vitamins.

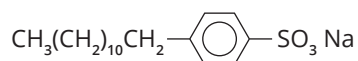
(2 marks)

b Explain why one of these vitamins is water soluble yet the other one is not.

(2 marks)

Question 20**(5 marks)**

The following is the formula of a detergent which is more useful than soap in areas of hard water.



- a** How does this detergent differ from soap? (1 mark)

- b** Is this detergent anionic, non-ionic or cationic? (1 mark)

- c** On the diagram, label the part of the molecule that is non-polar. (1 mark)

- d** Explain why detergents such as the above have an advantage over soap when used in hard water. (2 marks)

Question 21**(5 marks)**

Polytetrafluoroethene is a commonly used polymer with distinctive properties. A sample of polytetrafluoroethene has an average molar mass of 1500 kg mol^{-1} .

- a** Give the structure of the monomer that forms this polymer. (1 mark)

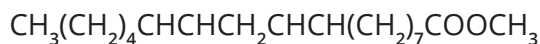
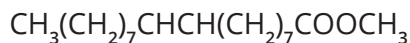
- b** Calculate the average number of monomer units in the polymer. (2 marks)

- c** Give one distinctive property of polytetrafluoroethene and a use related to that property. (2 marks)

Property
Use

Question 22**(9 marks)**

Biodiesel fuel is a mixture of esters obtained by reacting animal fats or vegetable oils with an alcohol in the presence of a catalyst. When a particular fat is reacted with an alcohol, a biodiesel containing the following esters is produced.



a Give the formula of the alcohol used in this reaction. (1 mark)

b i Give the structure and name of the other product of this reaction. (2 marks)

ii Briefly explain how the biodiesel can be separated from this product. (2 marks)

c Give the formula of an inorganic catalyst commonly used in the reaction. (1 mark)

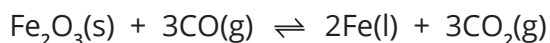
d Lipase, a biological catalyst, can also be used to catalyse the formation of a biodiesel from a fat or oil.

i To what class of compounds does lipase belong? (1 mark)

ii Give one advantage and one disadvantage of using lipase instead of an inorganic catalyst. (2 marks)

Question 23**(6 marks)**

Iron is produced by the reduction of iron ore in the blast furnace. One of the reactions that takes place is the reduction of iron(III) oxide by carbon monoxide as given by the following equation.



To produce iron, 350 tonnes of carbon monoxide is added to 500 tonnes of iron ore and allowed to react. This produces 320 tonnes of iron (1 tonne = 10^6 g).

- a** Determine which of the two reactants is the limiting reactant. (3 marks)

- b** Calculate the percentage yield of iron obtained in this reaction. (3 marks)

End of section two

Section three: Extended answer

32% (38 marks)

This section has 2 questions. You must answer **both** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three significant figures and include appropriate units where applicable.

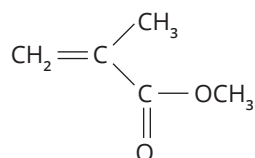
Do not use abbreviations, such as 'nr' for 'no reaction', without first defining them.

Suggested working time: 17 minutes

Question 24

(21 marks)

- a** Nylon is a synthetic polymer produced through a condensation polymerisation reaction.
- i** What is the essential difference between monomers that undergo condensation polymerisation and those that undergo addition polymerisation? (2 marks)
- _____
- _____
- _____
- ii** In terms of bonding between chains of nylon, explain why nylon can form very strong fibres. (1 mark)
- _____
- _____
- iii** The formula of methyl methacrylate is given below.



Methyl methacrylate is the monomer used to produce perspex, a transparent polymer used instead of glass in a number of applications. Deduce whether perspex is formed by addition or condensation polymerisation and draw a section of the polymer showing two monomer units. (2 marks)

Type of polymerisation	
Section of polymer showing 2 monomer unit	

- b** Proteins are natural polymers also produced through condensation polymerisation of amino acids. Identify, and give the structure of, the functional group common to both nylon and protein polymers. (1 mark)

- c** One particular α -amino acid that forms proteins consists of 34.3% carbon, 6.7% hydrogen, 45.7% oxygen and only one other element. Further analysis shows that 0.250 mol of this amino acid has a mass of 26.26 g.

- i** What other element is present in all amino acids? (1 mark)

- ii** Determine the molecular formula of this amino acid. (4 marks)

- iii** Deduce and draw the structure of this amino acid as it is most likely to exist in a solution with a pH of 7.

(2 marks)

- d i** The structure of the following amino acids is given in your data book: valine, cysteine, aspartic acid, leucine, lysine and threonine. These amino acids constitute part of most proteins and the interactions between their side groups contribute to the maintenance of the tertiary structure of the protein.

Identify the kind of interaction that exists between the side groups of each of the following pairs of amino acids. (4 marks)

Amino acids	Interaction (covalent/ionic/hydrogen bonds/dipole-dipole interaction/dispersion forces)
threonine and lysine	
cysteine and cysteine	
valine and leucine	
aspartic acid and lysine	

- ii Explain the effect of high temperature and extremes of pH on the interactions that maintain the tertiary structure of a protein. (4 marks)

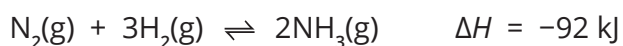
Effect of high temperature _____

Effect of extremes of pH _____

Question 25

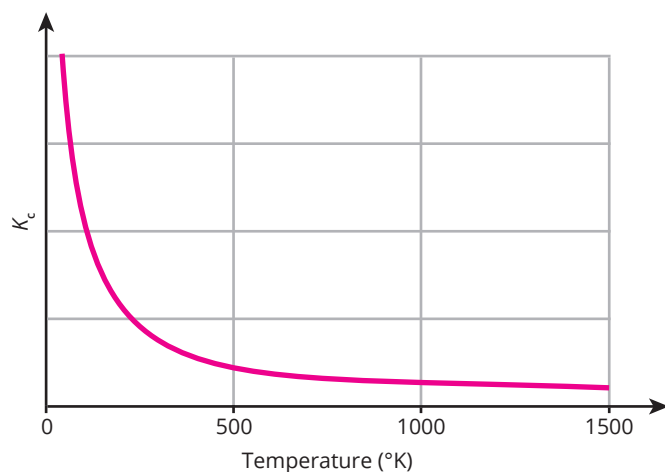
(17 marks)

Ammonia is one of our most important industrial chemicals. It is produced by the reaction of nitrogen and hydrogen gases according to the following equation.



- a i Write the equilibrium constant expression for this reaction. (1 mark)

- ii The graph below shows the change in the value of the equilibrium constant as the temperature increases.



Explain the relationship between the value to K_c and temperature in terms of equilibrium principles. (2 marks)

b Industrially, this process is carried out at 350°C to 550°C, at pressures between 100 atm and 250 atm, and with a porous iron oxide catalyst.

i Explain why the process is not carried out at a much higher or much lower temperature. (2 marks)

ii Why do elevated pressures increase both the rate of reaction and the yield of ammonia? (4 marks)

iii Explain why a catalyst increases the reaction rate and the advantage of having a catalyst that is porous. (3 marks)

c In the industrial production of ammonia, sufficient time is not allowed for the mixture to reach equilibrium. Instead, the gases leaving the converter are cooled. Ammonia is liquefied but not the unreacted nitrogen and hydrogen. These are then recycled into the converter to improve the overall yield of ammonia.

Explain why ammonia liquefies on cooling but nitrogen and hydrogen do not. (2 marks)

d Given that the ultimate percentage yield of ammonia by this method is 98%, calculate the volume of hydrogen gas, measured at 25°C and 100 kPa, that is required to produce 800 kg of ammonia. (3 marks)

End of questions